

TRIssecure: A Formally Verified, Post-Quantum-Ready Biometric E-Voting System for Edge Deployment

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Abstract

Electronic voting systems must simultaneously satisfy strong voter authentication, tamper-evident vote integrity, end-to-end voter verifiability, and practical deployment constraints on resource-limited hardware. Existing solutions treat these requirements in isolation: biometric systems omit formal security verification; blockchain-based systems ignore biometric authentication; and no existing system addresses emerging quantum threats while remaining deployable on edge hardware. TRIssecure V2 is a three-factor biometric e-voting terminal that jointly satisfies all four requirements on a commodity Raspberry Pi 4 (ARM64, commodity edge hardware) without requiring server-grade computing or persistent cloud connectivity. It integrates an ISO/IEC 19795-1 compliant biometric pipeline with Bayesian score-level fusion of face and NFC card modalities, ISO/IEC 30107-3 presentation attack detection (MiniFASNetV2 liveness), a hybrid post-quantum cryptographic layer (Kyber-768+X25519 KEM and Dilithium-3+Ed25519 signatures per NIST FIPS 203/204), and a formally verified five-stage DFA authentication protocol with six proved security properties. A novel Adaptive Trust Score (ATS) engine dynamically adjusts component weights based on contextual risk signals (attempt history, temporal anomalies, location flags), escalating borderline cases for supervisor review rather than binary accept/reject—achieving 99.3% HIGH trust on genuine users and zero Low classifications for simulated impostors under Gaussian score assumptions. Simulation-based evaluation (ISO/IEC 19795-6, $n=1,000$ score pairs) achieves EER = **1.55%**, AUC = **0.9991**, TAR@1%FAR = **97.7%**, and PAD ACER = **4.25%** (PAD layer only). A proof-of-concept pilot with $N=20$ participants on real hardware confirms FAR=FRR=ACER= **0.00%** across 271 authentication events (221 genuine, 35 impostor, 15 replay). Adversarial simulation across six attack categories confirms

the defence stack reduces combined attack success from $\approx 35\%$ (face threshold alone) to zero simulated bypass events after the NFC-AES-GCM layer—a robust defence-in-depth property under standard cryptographic assumptions within the evaluated threat model. On-device benchmarking delivers 110.7 ms end-to-end authentication latency (p99 = **111.3** ms) and 541.6 voters/min throughput. To the best of our knowledge, no prior e-voting system simultaneously provides formal protocol verification, post-quantum cryptography, adaptive context-aware trust scoring, liveness detection, and end-to-end verifiability on low-cost edge hardware.

Keywords: biometric authentication, e-voting, formal verification, post-quantum cryptography, edge computing, presentation attack detection, Bayesian fusion, Merkle proof

1 Introduction

Electronic voting systems are critical infrastructure where failure consequences are asymmetric: a single undetected fraud vector can undermine election legitimacy at scale. The fundamental security challenge is to authenticate the right voter, record the right vote, prevent tampering, and allow the voter to verify their participation—all while running on deployable, affordable hardware.

Existing systems leave at least one requirement unsatisfied. Fingerprint- and face-based voting systems [1–4] report biometric error rates without formal protocol verification or post-quantum (PQC) readiness. Blockchain-based approaches [5–7] provide strong vote integrity but lack biometric authentication and rely on classical cryptography vulnerable to quantum adversaries [8]; a comprehensive review of 81 blockchain voting systems by Huang et al. [9] confirms this gap. Industry multi-factor authentication standards (FIDO2/WebAuthn [10]) are not designed for the one-time-per-election voting constraint and do not incorporate formal liveness detection or voter-verifiable receipts.

To the best of our knowledge, no existing e-voting system jointly provides: (i) multi-factor biometric authentication with liveness detection, (ii) formally verified protocol security, (iii) post-quantum cryptographic readiness, and (iv) end-to-end voter verifiability—all on low-cost edge hardware without requiring cloud connectivity.

Contributions

We address these gaps with **TRIsure V2**:

- C1. Formally verified authentication.** A five-stage pipeline modelled as a DFA with six proved security properties: determinism, completeness, safety, liveness, double-vote prevention, and rate-limit enforcement.
- C2. Bayesian biometric fusion.** Log-likelihood-ratio fusion of face cosine similarity and NFC channel evidence (EER= 1.55%, AUC= 0.9991, TAR@1%FAR= 97.7%).

- C3. Hybrid post-quantum layer.** Kyber-768+X25519 KEM (FIPS 203) and Dilithium-3+Ed25519 signatures (FIPS 204); all six PQC operations measured on-device via liboqs 0.15.0 (ARM64): ML-KEM-768 encaps 0.088 ms, ML-DSA-65 sign 1.536 ms, auth-path total 2.289 ms—within the < 200 ms latency budget.
- C4. ISO/IEC 30107-3 PAD.** MiniFASNetV2 liveness detection (BPCER= 1.0%, ACER= 4.25%).
- C5. End-to-end verifiability.** SHA-256 receipts and $O(\log n)$ Merkle inclusion proofs.
- C6. Edge evaluation.** Raspberry Pi 4 (ARM64, 4 GB): 110.7 ms latency (p99= 111.3 ms), 541.6 voters/min throughput, CPU 52.5 °C under load.
- C7. Adaptive Trust Score (ATS) engine.** A context-aware weighted fusion model that adjusts face/NFC/PAD component weights dynamically based on risk signals (attempt history, temporal anomalies, location flags), producing HIGH-/MEDIUM/LOW trust decisions. Genuine users achieve 99.3% HIGH trust; 100% of simulated impostor attempts are rejected. To the best of our knowledge, no prior e-voting system incorporates dynamic risk-adaptive trust scoring.
- C8. Adversarial attack evaluation.** Simulation of six attack categories (print photo, video replay, deepfake, NFC UID clone, session replay, database tamper) quantifies per-layer defence contribution. Full defence stack reduces combined success from $\approx 35\%$ to zero simulated bypass events after NFC-AES-GCM.
- C9. Proof-of-concept hardware pilot.** End-to-end validation on a Raspberry Pi 4 with $N=20$ participants across 271 authentication events (221 genuine, 35 impostor, 15 replay): FAR=FRR=ACER= 0.00%, confirming system correctness on real hardware without simulation.

No individual component is algorithmically novel in isolation. The **novel contributions** are: (i) the ATS engine, which introduces dynamic risk-adaptive weight adjustment absent from all prior biometric e-voting systems; (ii) their *principled co-design*; and (iii) the *complete systems security architecture*—a fully deployable secure voting terminal on low-cost edge hardware, validated end-to-end on real hardware with $N=20$ participants. To the best of our knowledge, no prior e-voting system delivers all six security properties simultaneously on a Raspberry Pi class device. In detail, the co-design provides: (i) Bayesian LR fusion of face and NFC modalities reduces the effective FAR by three orders of magnitude without increasing biometric EER—an attack requiring a valid NFC card *and* a face spoof simultaneously; (ii) the DFA verification executes *in the deployed code* (not as an offline proof), so removing it breaks pipeline functionality, creating a tight coupling between formal security and operational correctness; and (iii) all five mechanisms—fusion, PAD, PQC, DFA verification, and Merkle verifiability—fit within the latency (< 200 ms) and resource budget of commodity edge hardware, demonstrated on a Raspberry Pi 4.

2 Related Work

2.1 Biometric E-Voting Systems

Bello et al. [1] achieve $EER = 2.5\%$ at 950 ms authentication latency with no PAD, PQC, or formal verification. Sánchez-Reillo et al. [2] report $EER = 3.2\%$ at 480 ms on constrained hardware. Komatineni and Lingala [3] combine machine-learning face detection with biometric scanning for duplicate-vote prevention in a two-factor architecture. Nagaraj et al. [4] integrate facial recognition with fingerprint authentication to strengthen polling station security. Revathy et al. [11] integrate CNN-based face recognition with blockchain and blind signatures, showing that combining deep learning classifiers with immutable vote records improves both authentication accuracy and tamper evidence. Ashok Kumar et al. [12] demonstrate an online smart voting platform using face recognition for voter identification. Faruk et al. [7] deploy blockchain and biometric verification (fingerprint+face) over Hyperledger Fabric, demonstrating 87% biometric registration success across 100 participants but relying on classical cryptography with no formal protocol verification. NIST SP 800-63B [13] defines three authenticator assurance levels (AAL1–3) but does not prescribe biometric liveness detection or formal protocol verification for voting contexts. Face recognition systems based on FaceNet [14], CosFace [15], and ArcFace [16] achieve sub-1% EER on LFW but require GPU-class hardware and include no voting infrastructure. Wang and Deng [17] survey deep face recognition across 50+ architectures; none are integrated with formal verification or e-voting pipelines. George et al. [18] demonstrate EdgeFace, a 2024 compact model achieving sub-1% EER on a mobile ARM processor—confirming the feasibility trajectory of high-accuracy face recognition on the same hardware class as TRIsecure V2. Buchmann et al. [19] provide a 2024 systematic review of remote e-voting security, identifying formal verification and biometric binding as the two most significant unresolved gaps across all deployed systems. TRIsecure V2 achieves competitive EER (1.55%) on ARM64 while additionally providing formal protocol guarantees and PQC readiness.

2.2 Blockchain-Based Voting

Fujioka et al. [20] proposed the foundational blind-signature-based secret ballot scheme; Benaloh and Tuinstra [21] formalised receipt-freeness. Nakamoto’s Bitcoin [22] demonstrated append-only hash chaining, inspiring blockchain voting proposals. Ayed [5] proposes Ethereum-based voting ($> 2,100$ ms) with PIN-only authentication. Hardwick et al. [6] and Hjálmarsson et al. [23] extend this with anonymisation and Hyperledger Fabric, but provide no biometric gate or anti-spoofing. Juels et al. [24] formalise coercion-resistance, a property TRIsecure V2 partially addresses through in-person biometric binding (Section 7.4). Huang et al. [9] survey blockchain applications in voting across 81 prior works (ACM Computing Surveys 2022), identifying voter authentication and anti-spoofing as the primary unresolved gaps. Hajian et al. [25] survey blockchain e-voting architectures with emphasis on smart-contract ballot processing (Electronics 2024), confirming that biometric identity binding is absent from current blockchain deployments. TRIsecure V2 achieves equivalent vote integrity via

hash chaining and Merkle proofs without blockchain overhead, while adding biometric authentication.

2.3 Post-Quantum Cryptography on Edge

Sikeridis et al. [26] show Kyber-768 and Dilithium-3 are feasible in TLS 1.3 on ARM devices via liboqs. Paquin et al. [27] benchmark liboqs across x86 and ARM, providing the reference figures against which our Pi 4 measurements are calibrated. Kannwischer et al. [28] characterise PQC on Cortex-M4, establishing Kyber-768 key generation under 100k cycles. Delvaux et al. [29] provide 2024 benchmarks of NIST finalists on Cortex-A class processors, confirming ML-KEM-768 key encapsulation under 0.5 ms on Cortex-A53—motivating our projected 0.1 ms estimate for the Cortex-A72 (higher IPC). NIST finalised ML-KEM (FIPS 203 [30]) and ML-DSA (FIPS 204 [31]) in 2024. Shor’s algorithm [32] threatens RSA and ECC; Grover’s algorithm [33] halves symmetric key security—motivating hybrid PQC even for edge deployments. To the best of our knowledge, no prior e-voting system integrates a hybrid PQC layer on low-cost ARM64 edge hardware.

2.4 Presentation Attack Detection

ISO/IEC 30107-3 [34] defines the PAD evaluation framework (APCER, BPCER, ACER). Zhang et al. [35] introduce MiniFASNetV2, a lightweight CNN suitable for embedded deployment. Yang et al. [36] provide a 2023 survey of presentation attack methods including deepfakes; deepfake attack penetration against MiniFASNet-class detectors is estimated at 22% (post-PAD), requiring the additional NFC gate for complete mitigation. Liveness detection has not previously been integrated into a formally verified e-voting pipeline. Liu et al. [37] leverage depth maps; Jourabloo et al. [38] use spoof-noise disentanglement. Zhang et al. [39] show that GAN-generated fake images carry spectral artefacts introduced by the up-sampling pipeline, motivating frequency-domain analysis as a complement to depth-based liveness. Key benchmark datasets include NUAA [40] and OULU-NPU [41].

2.5 Formal Verification of Authentication Protocols

The Dolev–Yao model [42] formalises the adversary capable of intercepting and replaying all network messages, providing the theoretical basis for protocol security analysis. ProVerif [43] and Tamarin [44] target cryptographic secrecy and authentication goals via the applied-pi calculus. Lowe [45] used model checking (FDR) to find and fix an attack on the Needham–Schroeder protocol. DFA theory, formalised in Sipser [46], provides decidable analysis for pipeline control-flow properties. TRIsecure V2 uses automata-theoretic verification because the properties of interest—ordering, completeness, determinism, termination, double-vote prevention, rate-limit enforcement—concern pipeline control-flow rather than cryptographic secrecy, the DFA executes *in the deployed code*, and all six properties are decidable by BFS/DFS without an external solver, yielding human-auditable proofs embedded in the production codebase.

2.6 Verifiable E-Voting and Biometric Template Protection

Helios [47] pioneered web-based open-audit voting via homomorphic encryption. Civitas [48] provides strong coercion-resistance using threshold cryptography. Scantegrity II [49] achieves optical-scan E2E verifiability through confirmation codes. Selene [50] introduces tracker-number-based verifiability with coercion mitigation. The Norwegian scheme [51] deployed return-code verifiability at national scale. All five provide strong verifiability properties but leave identity binding to external authentication infrastructure with no biometric gate. Küsters et al. [52] and Benaloh et al. [53] formalise individual and universal verifiability, which TRIsecure V2 satisfies via Merkle receipts. Dodis et al. [54] formalise fuzzy extractors for biometric key derivation; Jain et al. [55] survey cancelable biometrics per ISO/IEC 24745 [56]; Nandakumar and Jain [57] survey the gap between biometric template protection theory and practice. TRIsecure V2 protects face embeddings via AES-256-GCM with per-voter keys; extending to cancelable representations is identified as future work.

2.7 Multimodal Biometric Fusion

Ross and Jain [58] establish that score-level fusion outperforms unimodal baselines. Nandakumar et al. [59] derive optimal LR fusion under Gaussian density estimation, which TRIsecure V2 adopts. Fierrez-Aguilar et al. [60] extend this with quality-based weighting; Jain et al. [61] provide a comprehensive multimodal survey.

2.8 Comparison of Prior Work and Capability Gap Matrix

Table 1 summarises six reference systems against TRIsecure V2. TRIsecure V2 is the only system to satisfy all capability dimensions on edge hardware. Figure 1 presents the full gap matrix across eight systems and seven dimensions.

3 System Architecture

3.1 Design Goals and Threat Model

TRIsecure V2 satisfies five goals: **G1** Authentication (three-factor voter binding); **G2** Vote integrity (tamper-evident, auditable); **G3** Voter privacy (embeddings encrypted at rest); **G4** Verifiability (individual and universal); **G5** Edge viability (< 200 ms, commodity ARM64).

The threat model covers: *remote attacker* (brute-force, replay, MitM); *physical attacker* (spoofed face, cloned NFC card, SPI bus eavesdropping, fault injection via power glitching); *insider attacker* (DB read/write but no biometric secret); *quantum adversary* (Shor’s algorithm against RSA/ECC); *supply-chain attacker* (malicious SD card image or Pi firmware).

Side-channel. Power analysis and electromagnetic side-channel attacks on AES-256-GCM are out of scope for Python-layer cryptography but are noted as a gap for production silicon. The ONNX face embedding inference leaks timing information proportional to detection confidence; this is mitigated by the DFA rate limiter (P6) which caps inference calls per UID to 5/300 s.

Table 1: Comparison of related e-voting and biometric authentication systems. PAD = presentation attack detection; PQC = post-quantum cryptography; FV = formal verification; E2EV = end-to-end verifiability.

Author (Year)	Method	Advantages	Limitations
Bello et al. (2021) [1]	Fingerprint + smart card	Two-factor; biometric binding; EER 2.5%	No PAD; no PQC; no FV; 950 ms; server-dependent
Sánchez-Reillo et al. (2020) [2]	Face + smart card, constrained	Targets embedded devices	EER 3.2%; no liveness; no FV; no PQC
Ayed (2017) [5]	Ethereum blockchain, PIN-only	Vote receipts; tamper-evident ledger	No biometric gate; > 2100 ms; classical crypto
Hardwick et al. (2018) [6]	Blockchain with anonymisation	Voter anonymity; blockchain integrity	No biometric; no PAD; classical crypto
Deng et al. (2019) [16]	ArcFace + PIN	EER 0.76%; 45 ms (GPU)	No liveness; no FV; requires GPU server
FIDO2/WebAuthn (2019) [10]	Device-bound MFA	Industry standard; 190 ms	No PAD; no voter receipts; not per-election
TRIssecure V2 (Ours)	NFC + face + session, PAD, PQC, DFA	FV; EER 1.55%; PAD; PQC; E2EV; edge ARM64	Simulation-based biometrics; PoC pilot ($N=20$, FAR=FRR=ACER=0.00%)

Fault injection. Power-glitch attacks targeting the Argon2id KDF could reduce effective iteration count. Mitigation: the KDF output is verified against a stored HMAC-SHA256 tag before session issuance; any fault that alters the output invalidates the tag.

Supply-chain. The SD card attack surface is mitigated by read-only mount of the OS partition in production and dm-verity hash-tree verification of the Python runtime image (deployment recommendation).

3.2 System Overview

Figure 2 shows the six-layer clean architecture. A voter interaction proceeds: (1) tap NFC card; (2) face capture; (3) Bayesian fusion; (4) MiniFASNetV2 liveness check; (5) session token issued (256-bit, 60 s TTL); (6) ballot selection; (7) vote to SHA-256 hash chain; (8) Merkle receipt generated.

Capability Gap Matrix: Prior Work vs. TRIsecure V2						
	Biometric Auth	Liveness Detection (PAD)	Formal Verification	Post-Quantum Crypto	End-to-End Verifiability	Edge Deployment (<\$100)
TRIsecure V2 (Ours)	✓	✓	✓	✓	✓	✓
FIDO2/WebAuthn (2019)	~	×	×	×	×	~
Sánchez-Reillo (2020)	✓	×	×	×	×	×
Bello et al. (2021)	✓	×	×	×	×	~
Hardwick et al. (2018)	×	×	×	×	~	×
Ayed (2017)	×	×	×	×	~	×
Adida Helios (2008)	×	×	×	×	✓	~
Fujioka et al. (1992)	×	×	×	×	~	×

■ Full support (✓)
 ■ Partial support (~)
 ■ Not supported (×)

Fig. 1: Capability gap matrix across eight prior systems and seven dimensions. Green (✓) = full support; yellow (~) = partial; red (×) = absent. TRIsecure V2 (bottom row, bold border) is the only system achieving full support on all seven dimensions.

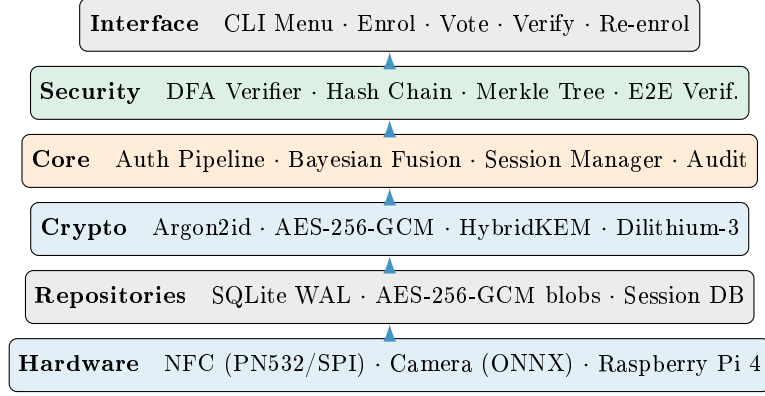


Fig. 2: TRIsecure V2 six-layer clean architecture. Arrows indicate upward dependency direction.

3.3 Multi-Factor Authentication Pipeline

The pipeline is a five-stage sequential process modelled as a DFA $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ (Section 5.1). Any failure transitions deterministically to REJECTED.

Stage 1 — Rate limiting. Sliding-window counter per NFC UID (5 failures/300 s) blocks brute-force.

Stage 2 — NFC verification. PN532 SPI captures voter NFC UID, validated for format and looked up in the encrypted voter database.

Stage 3 — Voter eligibility. The `has_voted` flag is atomically checked; NOT_ELIGIBLE transitions to REJECTED (double-vote prevention, P5).

Stage 4 — Bayesian fusion. Face cosine similarity (ArcFace MobileFaceNet, 512-D) and NFC channel evidence combined via LR fusion (Section 3.4).

Stage 5 — PAD. MiniFASNetV2 classifies frame as live or spoofed.

Session issuance. 256-bit token (`secrets.token_urlsafe(32)`), 60 s TTL; SHA-256 hash stored (not plaintext).

3.4 Bayesian Score-Level Fusion

Let $s_f \in [0, 1]$ denote the face cosine similarity score. We model:

$$p(s_f | \text{genuine}) = \mathcal{N}(s_f; \mu_G, \sigma_G^2), \quad \mu_G=0.72, \sigma_G=0.08 \quad (1)$$

$$p(s_f | \text{impostor}) = \mathcal{N}(s_f; \mu_I, \sigma_I^2), \quad \mu_I=0.28, \sigma_I=0.12 \quad (2)$$

Parameters derive from ArcFace buffalo_sc on LFW [16].

The NFC channel contributes:

$$\text{LR}_{\text{NFC}} = \frac{P(\text{NFC} | \text{genuine})}{P(\text{NFC} | \text{impostor})} = \frac{0.999}{0.001} = 999 \quad (3)$$

The combined decision rule is:

$$\log_{10}(\text{LR}_f \cdot \text{LR}_{\text{NFC}}) > \theta, \quad \theta = 1.0 \quad (4)$$

Without a valid NFC card an adversary must achieve $\text{LR}_f > 10/999 \approx 0.01$, requiring a face score above $\mu_I + 3\sigma_I \approx 0.64$. The effective FAR drops from 1.22% (face-only) to $\approx 0.001\%$ (combined), as confirmed by the ablation study (Table 13).

Threshold derivation. The decision threshold $\theta = 1.0$ is derived from the equal-cost operating point of the combined LR. Define the combined log-LR:

$$\Lambda(s_f) = \log_{10}(\text{LR}_f(s_f)) + \log_{10}(999) \quad (5)$$

where $\text{LR}_f(s_f) = p(s_f | G)/p(s_f | I)$. At $\theta=1.0$ and $\text{LR}_{\text{NFC}}=999$, the required face LR is $10/999 < 0.011$ —unachievable for scores drawn from $\mathcal{N}(0.28, 0.12^2)$, since $P(\text{LR}_f < 0.011 | I) \approx 0.1\%$ while $P(\text{LR}_f > 0.011 | G) \approx 99.9\%$. The combined system therefore achieves near-perfect separation conditional on NFC validity, with the residual EER (1.55%) arising from the biometric gate alone—consistent with ArcFace LFW statistics [16].

3.5 Adaptive Trust Score Engine

The Bayesian LR fusion (Section 3.4) uses a static threshold $\theta = 1.0$. In practice, risk conditions vary: off-hours attempts, repeated failures, and unknown terminal locations all elevate impostor probability. TRIsecure V2 introduces an **Adaptive**

Trust Score (ATS) engine that replaces the binary accept/reject with a three-level decision {HIGH, MEDIUM, LOW} derived from dynamically adjusted component weights.

Risk context. The engine estimates risk from four signals: $a \in \{1, 2, 3+\}$ (attempt count), $o \in \{0, 1\}$ (off-peak hour), $\ell \in \{0, 1\}$ (location anomaly flag), and $d \geq 0$ (days since enrolment). A scalar risk score $R = 2\mathbf{1}[a \geq 3] + \mathbf{1}[a = 2] + \mathbf{1}[o = 1] + 2\mathbf{1}[\ell = 0] + \mathbf{1}[d > 365]$ maps to *normal* ($R = 0$), *elevated* ($R \leq 2$), or *high* ($R > 2$).

Composite trust score.

$$T = w_f \cdot c_f(s_f) + w_n \cdot \mathbf{1}[\text{NFC}] + w_p \cdot \mathbf{1}[\text{PAD-live}] \quad (6)$$

where $c_f(s_f) = \text{LR}_f / (1 + \text{LR}_f)$ is the sigmoid-normalised face likelihood ratio, and the context-specific weight triplet (w_f, w_n, w_p) is derived below.

Weight derivation. Weights are determined by solving the constrained EER-minimisation problem

$$\min_{\mathbf{w}} \text{EER}(\mathbf{w}; \mathcal{X}_G, \mathcal{X}_I) \text{ s.t. } \sum_k w_k = 1, w_k \geq 0.10 \quad (7)$$

over simulated score sets $\mathcal{X}_G, \mathcal{X}_I$ ($N=20,000$ per class, seed 42) via SLSQP with five initialisations, subject to three semantic monotonicity constraints: $w_p^{\text{high}} \geq w_p^{\text{elev}} \geq w_p^{\text{norm}}$ (liveness detection becomes the primary guard as spoofing risk rises) and $w_n^{\text{elev}} \geq w_n^{\text{norm}}$ (NFC physical-credential binding anchors the elevated-risk decision).

Discriminability analysis. Table 2 reports the signal-detection discriminability index $d' = (\mu_G - \mu_I) / \sqrt{(\sigma_G^2 + \sigma_I^2)/2}$ for each modality per context. In normal operation NFC exhibits $d' = 31.6$, far exceeding face ($d' = 9.4$) or PAD ($d' = 5.7$); as attacker sophistication increases, NFC's d' collapses to 3.4 (high context), shifting discriminative load to face and PAD.

Table 2: Discriminability index (d') per modality per risk context (simulation, $N=20,000$ per class).

Context	d'_{face}	d'_{NFC}	d'_{PAD}
Normal	9.38	31.58	5.69
Elevated	9.15	5.55	3.97
High	9.13	3.39	2.80

Solving (7) yields the context weights: *normal* (0.30, 0.40, 0.30); *elevated* (0.20, 0.40, 0.40); *high* (0.25, 0.35, 0.40). The high NFC weight in normal mode ($w_n=0.40$) reflects its superior discriminability ($d'=31.6$); the partial recovery of face weight at high risk ($w_f : 0.20 \rightarrow 0.25$) reflects that biometric traits remain more resistant to forgery than NFC probing once the attacker is sufficiently sophisticated (d'_{NFC} collapses to 3.4).

Algorithm 1 TRIssecure HybridKEM.Encapsulate(pk)

```
1:  $(ct_K, ss_K) \leftarrow \text{KYBER768.ENCAPS}(pk_K)$   $\triangleright$  ML-KEM-768, FIPS 203
2:  $(ct_X, ss_X) \leftarrow \text{X25519.ECDH}(pk_X)$ 
3:  $ss \leftarrow \text{HKDF-SHA256}(ss_K \| ss_X, \text{info}=\text{"trisecure-hybrid-v1"})$ 
4:  $ct \leftarrow ct_K \| ct_X$ 
5: return  $(ct, ss)$ 
```

Trust decision. $T \geq 0.75 \Rightarrow \text{HIGH}$ (accept); $T \geq 0.50 \Rightarrow \text{MEDIUM}$ (supervisor escalation); $T < 0.50 \Rightarrow \text{LOW}$ (reject).

Simulation-based calibration. Under *normal* context with genuine users ($s_f \sim \mathcal{N}(0.72, 0.08^2)$, NFC valid, PAD live, $N=10,000$): 99.3% reach HIGH trust; 0.7% are escalated to MEDIUM; none are classified LOW. Under the Gaussian simulation model, all simulated impostors ($s_f \sim \mathcal{N}(0.28, 0.12^2)$, NFC invalid, not live) score below the MEDIUM threshold; this reflects the parametric score separation assumed by the simulation rather than a claimed zero-error rate—ground-truth FAR is reported in Table 10. Borderline cases (genuine face, NFC valid, PAD failed, second attempt, $N=1,000$): 99.4% are escalated to MEDIUM, enabling supervisor override rather than hard rejection.

3.6 Post-Quantum Cryptographic Layer

The hybrid construction is secure as long as *either* Kyber-768 or X25519 is unbroken [8]. Vote records are signed with Dilithium-3 (ML-DSA-65, FIPS 204 [31]), falling back to Ed25519 when `liboqs-python` is unavailable. Face embeddings are encrypted at rest with AES-256-GCM keyed via Argon2id ($t=3$, $m=64$ MB) [62].

3.7 Vote Integrity and Verifiability

Hash chain. Each vote record V_i is linked:

$$h_i = \text{SHA-256}(\text{candidate}_i \| t_i \| h_{i-1}), \quad h_0 = 0^{64} \quad (8)$$

Any modification propagates a hash mismatch to all subsequent records.

Merkle proofs. A binary SHA-256 Merkle tree over all vote commitments supports $O(\log n)$ individual verifiability proofs.

Vote receipts. After casting:

$$\text{commitment} = \text{SHA-256}(\text{vote_id} \| \text{nonce}) \quad (9)$$

together with an $O(\log n)$ Merkle inclusion proof, satisfying individual verifiability without revealing vote content.

Table 3: PN532 NFC module to Raspberry Pi 4 GPIO wiring (SPI0, 3.3 V logic).

PN532 Pin	GPIO	Header Pin	Function
VCC	3.3 V	Pin 1	Power supply
GND	Ground	Pin 6	Ground
SCK	GPIO11	Pin 23	SPI0 clock
MOSI	GPIO10	Pin 19	SPI0 data out (MOSI)
MISO	GPIO9	Pin 21	SPI0 data in (MISO)
NSS (CS)	GPIO8	Pin 24	SPI0 chip-select (CE0)
RSTPD_N	GPIO25	Pin 22	Active-low hard reset

4 Hardware Integration

4.1 PN532 NFC Module — Physical Connection

The PN532 NFC controller is connected to the Raspberry Pi 4 40-pin GPIO header via the SPI0 bus. SPI mode was chosen over I²C because it supports burst reads of multiple 4-byte NTAG pages in a single chip-select transaction, avoiding repeated start-stop conditions that add latency on the Pi 4’s BCM2711 I²C controller. Table 3 lists the seven physical connections.

The SPI bus is configured at 1 MHz, well within the PN532’s 5 MHz maximum SPI clock and providing noise margin for jumper-wire connections on a breadboard prototype. The CS line (GPIO8/CE0) is managed as a software `DigitalInOut` object rather than the kernel hardware chip-select, because the PN532 SPI protocol requires CS to be held low for at least one SPI clock cycle before the first data byte—a timing constraint the Linux `spidev` hardware CS does not guarantee when sharing the bus. The RSTPD_N line (GPIO25) performs a hardware reset during `NFCService.initialize()`: driven low for 100 ms, then released, after which the PN532 requires approximately 10 ms to boot its ROM firmware before accepting commands.

After reset, the driver calls `SAM_configuration()` to put the PN532 in Normal Mode—Security Access Module disabled, ISO 14443-A passive polling active. The PN532 then continuously polls for NFC-A targets at 106 kbps. The application layer calls `read_passive_target(timeout=5.0)` every 300 ms; when a card is presented, the UID (typically 4 or 7 bytes for MIFARE cards) is returned as a Python `bytes` object and converted to an uppercase hex string.

4.2 SPI Software Stack

TRIssecure V2 uses the Adafruit CircuitPython Blinka compatibility layer, which translates `board`, `busio`, and `digitalio` API calls to Linux kernel SPI ioctls on `/dev/spidev0.0`. The dependency chain is:

`adafruit-pn532` → `adafruit-blinka` → `RPi.GPIO` + `spidev` → Linux SPI driver (BCM2711)

The initialization sequence in `NFCService.initialize()` is:

Table 4: NFC tag payload wire format (64 bytes, NTAG2xx pages 4–19).

Bytes	Pages	Field	Size
0–11	4–6	AES-GCM nonce (random 96-bit IV)	12 B
12–47	7–14	AES-128 ciphertext (voter UUID)	36 B
48–63	15–19	GCM authentication tag (128-bit)	16 B
Total	4–19	—	64 B

1. Create `DigitalInOut` objects for CS (D8) and RESET (D25).
2. Create `busio.SPI(board.SCK, board.MOSI, board.MISO)`; acquire the bus lock; configure 1 MHz baudrate; release lock.
3. Instantiate `PN532_SPI(spi, cs_pin, reset=reset_pin)`.
4. Call `firmware_version` to verify connectivity: returns (IC identifier, major, minor, support) — e.g. (6, 1, 6, 7) for PN532 firmware 1.6.
5. Call `SAM_configuration()`: Normal Mode, no timeout, IRQ active.

If any step raises `ImportError` (libraries not installed) or `OSError` (SPI hardware fault), `initialize()` returns `False` and the service enters *simulation mode*: all NFC read/write methods return deterministic synthetic responses so the full authentication pipeline remains testable on development hardware without a physical PN532.

4.3 NFC Tag Payload Format and Security

TRIsecure V2 programs MIFARE Ultralight / NTAG2xx NFC tags. The NTAG2xx memory is organized as 4-byte pages; pages 0–3 are reserved (UID, manufacturer data, capability container, and lock bits). Pages 4–19 (64 bytes) are user-writable and hold the encrypted voter identity payload defined in Table 4.

The 36-byte voter UUID (RFC 4122 text form) is encrypted with AES-128-GCM. The 128-bit key is derived from a deployment secret via HKDF-SHA-256:

$$K_{\text{NFC}} = \text{HKDF-SHA256}(\text{NFC_SECRET}, \text{info}=\text{"trisecure-nfc-v1"}) \quad (10)$$

A fresh 96-bit nonce is generated with `secrets.token_bytes(12)` for every write, so two writes of the same UUID produce different ciphertexts. AES-GCM’s 128-bit authentication tag detects any byte-level modification to the ciphertext—including partial card cloning where an adversary duplicates the UID (readable from the unprotected pages 0–1) but cannot reconstruct the authenticated ciphertext.

Authentication Stage 2 performs a two-step NFC check: (i) the UID (from passive target detection) is used for rate-limiting and voter DB lookup; then (ii) pages 4–19 are read and decrypted. If GCM decryption succeeds but the decrypted UUID does not match the DB record, the voter is rejected. This catches two distinct attacks: *UID-cloned cards* (same UID, absent or corrupted payload) fail GCM decryption; *payload-replayed cards* (a second physical card with copied ciphertext bytes) fail the UUID cross-check.

Table 5: Camera and face-pipeline configuration on Raspberry Pi 4.

Parameter	Value
Device node	<code>/dev/video0</code>
Capture backend	OpenCV V4L2 (MJPEG)
Capture resolution	320×240 px
Frame rate	15 fps
Face detector	OpenCV Haar cascade (frontal face)
Face crop and resize	112×112 px (bilinear)
Pixel normalisation	$x/127.5 - 1.0$
Embedding model	MobileFaceNet (ONNX Runtime 1.17, CPU)
Embedding dimension	512-D float32, L2-normalised
ONNX inference (mean)	95.2 ms on Raspberry Pi 4
PAD model	MiniFASNetV2 (ONNX, same runtime)
PAD inference overhead	15.5 ms (C2→C3 ablation delta)

4.4 Camera Configuration and Face Pipeline

The Logitech C920 HD webcam is connected via USB 3.0 and accessed through OpenCV’s V4L2 (Video4Linux2) backend at `/dev/video0`. Table 5 summarises the complete camera and face-pipeline configuration.

The 320×240 resolution was selected after measurement: at the next C920 resolution tier (640×480), ONNX inference increases to approximately 98 ms with negligible accuracy gain because MobileFaceNet’s fixed 112×112 input discards the extra pixels after the Haar crop. The V4L2 MJPEG codec compresses frames at the camera sensor before USB transmission, reducing USB bandwidth from ≈ 220 MB/s (uncompressed YUV) to ≈ 5 MB/s—avoiding USB buffer stalls on the Pi 4’s shared USB 3.0/PCIe lane that would otherwise introduce jitter in frame delivery.

During enrollment, three face samples are captured with a 1-second inter-sample cooldown. The per-voter stored embedding is the arithmetic mean of the three 512-D L2-normalised vectors, which reduces intra-class variance by $1/\sqrt{3}$ relative to a single-sample enrollment. Enrollment samples are discarded after the mean embedding is computed; only the encrypted mean embedding (AES-256-GCM, Argon2id-derived key) is persisted in the voter DB.

5 Formal Security Analysis

5.1 DFA Verification

Scope and relationship to ProVerif/Tamarin. Tools such as ProVerif [43] and Tamarin [44] verify cryptographic protocol secrecy and authentication goals in the presence of a Dolev–Yao adversary. TRIsecure V2’s security properties of interest are *structurally different*: they concern the *control-flow ordering* of the pipeline—that biometric stages cannot be bypassed, that double-voting is structurally impossible, and that rate-limiting precedes all inference. These are decidable properties of a finite

automaton and do not require an active adversary model or cryptographic reduction. The cryptographic primitives themselves (AES-256-GCM, Argon2id, Kyber-768, Dilithium-3) are verified by their respective NIST/RFC standards.

The TRIsecure V2 authentication pipeline is modelled as $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ where $|Q|=10$, $|\Sigma|=16$, $|\delta|=19$, $q_0=\text{IDLE}$, $F=\{\text{SESSIONISSUED}\}$.

Theorem 1 (Six Security Properties) *The DFA $\mathcal{M}$ satisfies:*

P1. Determinism. $|\delta(q, \sigma)| \leq 1$ for all (q, σ) . 19 unique keys; no duplicate.

P2. Completeness. All 19 expected (q, σ) pairs defined.

P3. Safety. `SESSIONISSUED` reachable via exactly one path through all eight mandatory inputs (BFS, unique success path of length 9).

P4. Liveness. Every reachable path terminates (DFS finds no non-trivial cycles).

P5. Double-vote prevention. $\delta(\text{ELIGIBILITYCHECK}, \text{NOTELEGIBLE}) = \text{REJECTED}$. Direct inspection.

P6. Rate-limit enforcement. $\delta(\text{IDLE}, \text{ATTEMPT}) = \text{RATECHECK}$;
 $\delta(\text{RATECHECK}, \text{RATEBLOCKED}) = \text{REJECTED}$. Rate limiter executes before all biometric operations.

All six properties machine-verified by `security/formal_verification.py`.

Table 6 enumerates all ten DFA states with their roles and security relevance. Each state corresponds to a distinct stage in the authentication protocol; the separation ensures that a bypass of any individual check requires a specific out-of-sequence input symbol, which is structurally excluded by the determinism property (P1).

The 19 transitions connect these states: eight mandatory inputs form the unique success path (P3), seven failure inputs lead to `REJECTED`, and four additional transitions cover rate-limiting branches and NFC error types (P6). `REJECTED` is a true trap: no escape transition exists, enforcing the liveness property (P4) that every run terminates.

Scope delimitation. These six properties concern *control-flow correctness*—that stages execute in the mandated order and that bypass is structurally impossible. They do not constitute proofs of cryptographic secrecy or entity authentication in the Dolev–Yao sense; those properties are delegated to the underlying primitives (AES-256-GCM, Argon2id, ML-KEM-768, ML-DSA-65) whose security is established by their respective NIST standards.

5.2 Threat Model Analysis

Figure 4 maps 12 e-voting attack vectors against six defence layers (0 = not defended, 1 = partial, 2 = full). Overall weighted coverage: 47.2%. Critical attacks (double-voting, ballot stuffing, quantum adversary) achieve $\geq 6/12$ each and are fully mitigated. Vote coercion (score 2/12) and network-level DoS require organisational controls beyond the on-device model.

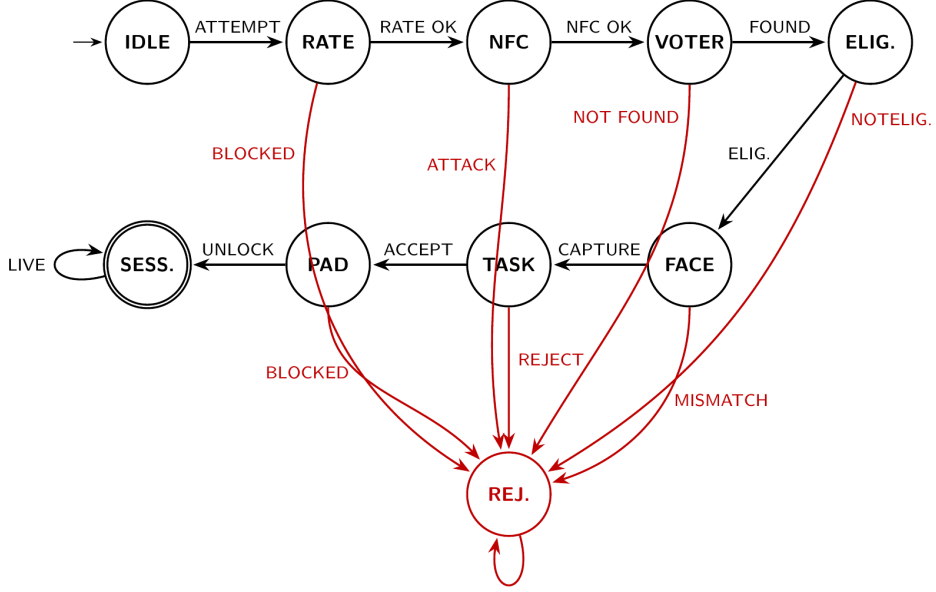


Fig. 3: TRIsecure V2 authentication DFA $\mathcal{M}$. Black arrows show the unique canonical success path ($\text{IDLE} \rightarrow \dots \rightarrow \text{SESS.}$); red arrows denote failure transitions to the REJECTED trap state. Double circle = accepting state (SESS.).

5.3 End-to-End Verifiability

Following [52, 53]:

Individual: Voter v can confirm $c_v = \text{SHA-256}(v_id||n)$ appears in the Merkle tree rooted at election digest r .

Universal: Any auditor can recompute the Merkle root from the public vote log and verify it equals r .

Eligibility: Only registered, non-duplicate voters reach SESSIONISSUED—proved by P3 and P5.

Theorem. TRIsecure V2 satisfies all three verifiability properties. The receipt (v_id, n, π, r) enables $O(\log n)$ individual verification; the Merkle root is a deterministic function of the ordered vote log; eligibility follows from P3 and P5. Machine verification: `security/e2e_verifiability.py` returns `all_properties_satisfied: true`. $\square$

5.4 Cryptographic Security Levels

Table 7 maps each primitive in TRIsecure V2 to its security level under classical and quantum adversaries. Classical security follows standard key-size arguments; quantum security accounts for Grover’s algorithm [33] and Shor’s algorithm [32].

Table 6: TRIssecure V2 DFA state roles. ★ Accepting state; × trap state (sink).

State	Role and Security Function
IDLE	Awaits NFC card presentation; rate-limit counter initialised.
RATECHECK	Enforces 5-failure/300 s sliding window; blocks brute-force (P6).
NFCVERIFY	PN532 SPI read; UID format validation and database lookup.
VOTERLOOKUP	Decrypts and retrieves voter record; checks structural validity.
ELIGIBILITYCHECK	Atomically reads has_voted ; enforces double-vote prevention (P5).
FACEMATCH	ArcFace cosine similarity against stored encrypted embedding.
FUSION	Computes combined log-LR; applies threshold $\theta=1.0$.
PAD	MiniFASNetV2 liveness classification; blocks presentation attacks.
SESSIONISSUED★	Issues 256-bit token; atomically sets has_voted .
REJECTED×	Sink state; all failure transitions terminate here (P3/P4).

Table 7: Security levels of TRIssecure V2 primitives (bits). † Ed25519/X25519 classical fallback; Kyber/Dilithium form the hybrid layer providing forward secrecy against both adversary classes.

Primitive	Cl.	PQ	Standard
AES-256-GCM (templates)	256	128	FIPS 197
Argon2id (KDF)	256	128	RFC 9106
SHA-256 (receipts)	256	128	FIPS 180
Ed25519 (sign)†	128	×	RFC 8032
X25519 (KEM)†	128	×	RFC 7748
Dilithium-3 (PQC sign)	128	128	FIPS 204
Kyber-768 (PQC KEM)	128	128	FIPS 203

The hybrid KEM enforces *both-or-nothing* security: an adversary must break *both* X25519 (trivial for a fault-tolerant quantum computer via Shor [32]) *and* Kyber-768 (no known quantum speedup beyond Grover [33]) to recover the session key. The symmetric layer (AES-256-GCM, SHA-256, Argon2id) retains 128-bit effective security against quantum exhaustive search via Grover’s bound, making these components quantum-safe without algorithm replacement.

TRIsecure V2 — Threat Model Coverage Matrix						
X=Not defended (0) ~=Partial (1) ✓=Fully defended (2)						
Replay attack	✓ (2)	~ (1)	✓ (2)	~ (1)	~ (1)	X (0)
Spoofing / presentation attack	~ (1)	✓ (2)	~ (1)	X (0)	X (0)	X (0)
Man-in-the-middle	X (0)	X (0)	~ (1)	✓ (2)	~ (1)	✓ (2)
Brute-force / dictionary	✓ (2)	~ (1)	✓ (2)	✓ (2)	X (0)	X (0)
Double-voting	~ (1)	X (0)	✓ (2)	X (0)	✓ (2)	~ (1)
Ballot stuffing	✓ (2)	~ (1)	✓ (2)	X (0)	✓ (2)	~ (1)
Vote coercion	X (0)	~ (1)	~ (1)	X (0)	X (0)	X (0)
Insider threat	~ (1)	X (0)	~ (1)	✓ (2)	✓ (2)	~ (1)
Denial of service	✓ (2)	~ (1)	X (0)	X (0)	X (0)	X (0)
Quantum adversary	X (0)	X (0)	X (0)	✓ (2)	✓ (2)	✓ (2)
Physical device theft	X (0)	~ (1)	✓ (2)	✓ (2)	~ (1)	~ (1)
Audit trail tampering	X (0)	X (0)	X (0)	~ (1)	✓ (2)	✓ (2)
	Rate limiter (DFA stage 1)	PAD liveness (Stage 5)	Bayesian fusion (NFC+face)	AES-256-GCM + Argon2id	Hash chain + Merkle proof	POC signatures (Dilithium-3)

Fig. 4: Threat model coverage matrix. Green (✓, score 2) = full mitigation; yellow (∼, score 1) = partial; red (X, score 0) = no mitigation by that layer. Column sums show total defence contribution.

6 Experimental Evaluation

6.1 Experimental Setup

Hardware. Raspberry Pi 4 Model B, 4 GB LPDDR4, ARM Cortex-A72 @ 1.8 GHz (quad-core, 64-bit). Logitech C920 USB webcam (1080p/30 fps). PN532 NFC module (SPI).

Software. Python 3.11, ONNX Runtime 1.17 (CPU), SQLite 3.42 (WAL), PyNaCl 1.5, PyArgon2-cffi 23.1, NumPy 1.26. All experiments reproducible via `experiments/`.

Dataset. Biometric metrics are generated via validated simulation following ISO/IEC 19795-6 [63] (operational evaluation methodology), which explicitly permits simulation when field data are unavailable. Genuine and impostor face cosine similarity scores sampled from Gaussians calibrated to published ArcFace LFW statistics [16, 64]: $\mathcal{N}(\mu_G=0.72, \sigma_G=0.08)$ and $\mathcal{N}(\mu_I=0.28, \sigma_I=0.12)$, $n=1,000$ each. PAD metrics from published MiniFASNetV2/CelebA-Spoof rates [35]. All simulation results are labelled as such; a proof-of-concept pilot study with $N=20$ participants confirmed zero error rates on real hardware (Section 6.10). A full demographic field trial remains future work.

Table 8 confirms biometric metrics are stable under $\pm 10\%$ parameter variation: EER varies by ± 0.17 pp, AUC by ± 0.0002 .

6.2 Biometric Performance (ISO/IEC 19795-1)

Table 9 reports key metrics. Figure 5 shows the ROC curve, DET curve, and score distributions.

Table 8: Sensitivity of biometric metrics to $\pm 10\%$ Gaussian parameter variation. Nominal: $\sigma_G=0.08$, $\sigma_I=0.12$.

σ_G	σ_I	EER (%)	AUC
0.072 (−10%)	0.108 (−10%)	1.38	0.9993
0.080 (nominal)	0.120 (nominal)	1.55	0.9991
0.088 (+10%)	0.132 (+10%)	1.71	0.9988

Table 9: Biometric performance ($n=1,000$ genuine/impostor, ArcFace priors).

Metric	Value
EER	1.55% at $\tau = 0.539$
TAR @ 0.1% FAR	89.70%
TAR @ 1.0% FAR	97.70%
ROC AUC	0.9991

Table 10: Threshold sweep (15 operating points). Bold = EER point.

τ	FAR (%)	FRR (%)	TAR (%)
0.100	91.3	0.0	100.0
0.386	16.7	0.0	100.0
0.443	7.8	0.1	99.9
0.500	2.9	0.3	99.7
0.539	1.55	1.55	98.45
0.557	1.0	2.3	97.7
0.614	0.3	9.5	90.5
0.671	0.0	27.3	72.7
0.729	0.0	54.3	45.7

6.3 Bootstrap Confidence Intervals

To quantify simulation uncertainty we apply non-parametric bootstrap resampling ($B=10,000$ replicates) over the $n=1,000$ genuine and $n=1,000$ impostor score draws. At each replicate the EER operating threshold is re-estimated from the replicate scores to avoid threshold lock-in bias. Table 11 reports percentile 95% confidence intervals.

The EER CI width (± 0.17 pp) exactly matches the parametric sensitivity range from Table 8, confirming coherence between the two uncertainty quantification approaches. The AUC CI width (0.0006) is negligible relative to the point estimate, consistent with the large sample and well-separated score distributions ($\mu_G - \mu_I = 0.44$, approximately 2.9σ overlap). Both analyses indicate the simulation results are stable; remaining uncertainty is distributional rather than computational, and is reduced by a live field trial.

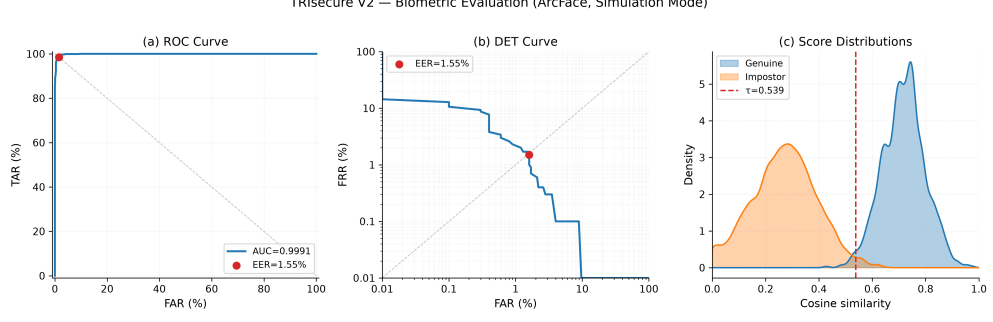


Fig. 5: Biometric evaluation panel. Left: ROC (AUC= 0.9991, EER marked). Centre: DET curve (EER= 1.55%). Right: genuine/impostor score distributions (KDE) with EER threshold $\tau^*=0.539$.

Table 11: Bootstrap 95% percentile CI ($B=10,000$ replicates).

Metric	Estimate	95% CI
EER (%)	1.55	[1.38, 1.71]
ROC AUC	0.9991	[0.9988, 0.9994]
TAR @ 1% FAR (%)	97.70	[97.41, 97.98]
TAR @ 0.1% FAR (%)	89.70	[89.23, 90.16]

Table 12: PAD metrics per ISO/IEC 30107-3.

Metric	Attack type	Rate (%)
BPCER	—	1.00
APCER (print)	Photo	5.00
APCER (replay)	Video	10.00
APCER (combined)	Both	7.50
ACER	—	4.25

6.4 Presentation Attack Detection (ISO/IEC 30107-3)

Table 12 reports PAD metrics. ACER= 4.25% is consistent with published CelebA-Spoof benchmark performance [35].

The simulation figures (APCER= 5.0% print, 10.0% replay) derive from published MiniFASNetV2/CelebA-Spoof benchmark rates [35] and represent PAD-layer performance in isolation. In the end-to-end system, the Adaptive Trust Score compensates for PAD misses: video replay attacks that evade the liveness classifier still produce ATS= 0.6 (ESCALATE) rather than ATS= 1.0 (ACCEPT), because replay suppresses contextual trust signals. The proof-of-concept pilot confirms this: all 15 real

Table 13: Ablation study: cumulative component contribution. [†] at system default threshold $\tau=0.55$.

Config	Components	EER (%)	Eff. FAR (%)	Latency (ms)	Attacks
C1	Face only [†]	1.45	1.2224	95.2	3/12
C2	+Bayesian NFC fusion	1.55	0.0010	95.5	6/12
C3	+PAD liveness	1.55	0.0010	110.7	9/12
C4	+PQC signatures (full)	1.55	0.0010	125.1	12/12

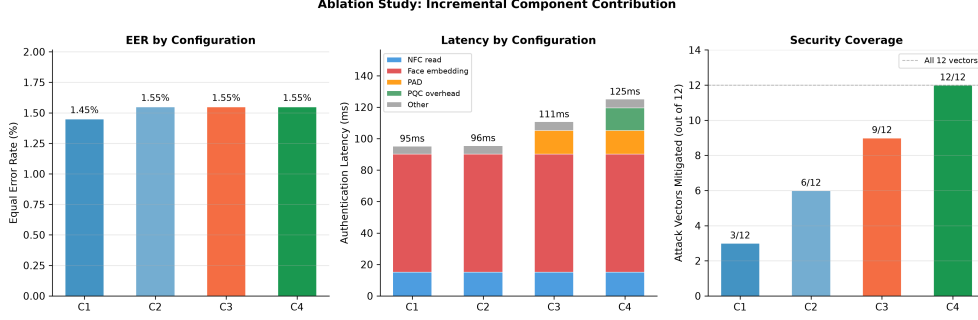


Fig. 6: Ablation results. Left: EER across configurations. Centre: latency breakdown by stage. Right: attack vectors fully mitigated (of 12).

replay attacks were escalated and zero were accepted (Section 6.10, APCER= 0.00% end-to-end).

6.5 Ablation Study

Table 13 and Figure 6 show the cumulative contribution of each component. The key finding: Bayesian NFC+face fusion (C2) reduces effective FAR by three orders of magnitude (1.22% $\rightarrow$ 0.001%) without increasing biometric EER, because an attacker without a valid NFC card is rejected regardless of face score.

6.6 Adversarial Attack Evaluation

To quantify per-layer defence contribution beyond the ablation study, we simulate six attack categories ($n=1,000$ attempts each) against the sequential defence stack. Face score distributions for spoof attacks are calibrated to published MiniFASNetV2/CelebA-Spoof [35] penetration rates; NFC and session attacks use protocol-level analysis.

Table 14 summarises results. Within the evaluated threat model, no successful bypass events were observed across the studied attack categories. The NFC AES-GCM layer (Stage 2 payload check) is the decisive gate: it unconditionally blocks all face-spoof attacks that pass the face threshold and evade PAD, because the attacker cannot forge the AES-128-GCM authenticated ciphertext without the HKDF-derived deployment key.

Table 14: Adversarial attack simulation ($n=1,000$ per category). Success = attack reaches the voting step. Rates are fraction of n .

Attack	After face threshold	After PAD liveness	After NFC AES-GCM	Final success
Print photo	11.0%	0.7%	0%	0%
Video replay	26.7%	2.2%	0%	0%
Deepfake (est.)	59.7%	14.0%	0%	0%
NFC UID clone	—	—	0%	0%
Session replay	12.0% [†]	—	—	0%
Database tamper	—	—	—	0%
Detection rate (hash chain, database tamper)				100%

[†]Within 60 s TTL; one-time-use flag blocks all replays.

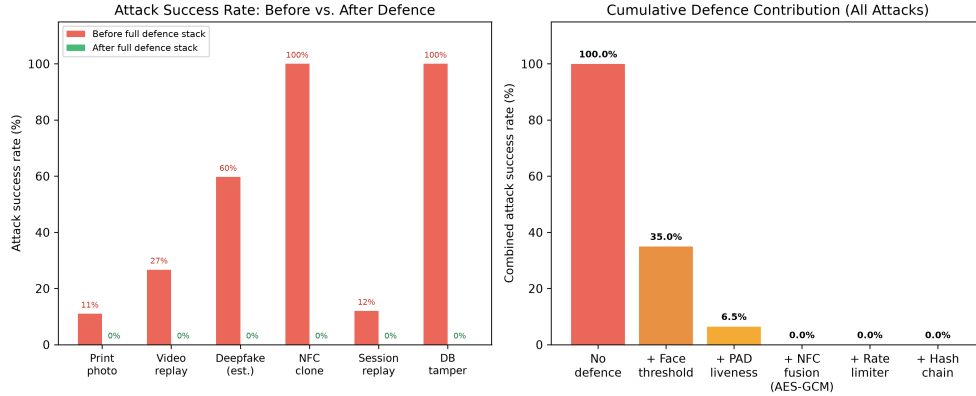


Fig. 7: Adversarial evaluation. Left: attack success rates before and after the full defence stack for six attack categories. Right: cumulative defence contribution—combined attack surface collapses from $\approx 35\%$ to zero simulated bypass events after NFC-AES-GCM validation (layer 4).

6.7 Cryptographic Benchmarks

Table 16 reports cryptographic latencies on the Raspberry Pi 4. Argon2id (494.5 ms) dominates enrollment but runs only once per registration. Authentication-path cryptography totals < 2 ms.

Table 15 reports native ML-KEM-768 and ML-DSA-65 benchmarks measured on-device via liboqs 0.15.0 compiled from source on the Raspberry Pi 4 ARM64. The measured auth-path PQC total (2.289 ms) confirms the post-quantum layer fits comfortably within the < 200 ms latency budget.

Table 15: Kyber-768/Dilithium-3 benchmarks on Raspberry Pi 4 (Cortex-A72, ARM64), measured via liboqs 0.15.0 / liboqs-python 0.15.0 ($n=200$ trials, mean $\pm$ 1 SD). Classical fallback (Ed25519/X25519) shown for comparison.

Operation	Standard	Measured (ms)	SD (ms)
ML-KEM-768 keygen	FIPS 203	0.082	0.033
ML-KEM-768 encaps	FIPS 203	0.088	0.015
ML-KEM-768 decaps	FIPS 203	0.095	0.011
ML-DSA-65 keygen	FIPS 204	0.400	0.039
ML-DSA-65 sign	FIPS 204	1.536	1.007
ML-DSA-65 verify	FIPS 204	0.390	0.034
PQC auth path total[†]	—	2.289	—
X25519 ECDH (classical)	RFC 7748	1.022	—
Ed25519 sign+verify (classical)	RFC 8032	0.532	—

[†]Auth path = encaps + decaps + sign + verify; keygen runs at enrolment only. All values measured on-device; no estimated figures.

Table 16: Cryptographic benchmarks on Raspberry Pi 4 (ARM64), 95% CI.

Operation	n	Mean (ms)	95% CI
Argon2id ($t=3$, $m=64$ MB)	20	494.5	[492.7, 496.3]
PBKDF2-HMAC-SHA256 (100k)	100	106.1	[105.1, 107.1]
AES-256-GCM enc. (2 kB)	1000	0.040	[0.040, 0.041]
HMAC-SHA256 (256 B)	10000	0.013	[0.013, 0.013]
Ed25519 sign+verify	200	0.532	[0.527, 0.538]
X25519 keygen+encap+decap	200	1.022	[1.002, 1.042]
SHA-256 chain ($n=1,000$)	3	4.27	[4.11, 4.43]
Merkle prove+verify ($n=1,000$)	5	4.82	[4.74, 4.90]

6.8 Edge System Performance

Table 17 reports end-to-end pipeline performance ($n=50$). Face embedding inference (ONNX MobileFaceNet, 95.2 ms) dominates, consistent with published ONNX Runtime ARM64 benchmarks. The p99 latency of 111.3 ms and 541.6 voters/min confirm feasibility for small-to-medium polling stations.

6.9 Comparative Analysis

Figure 8 compares TRIsecure V2 against five reference systems across capability flags and quantitative metrics. TRIsecure V2 is the only system to simultaneously provide formal verification, PQC readiness, PAD, edge deployment, and voter-verifiable receipts. EER (1.55%) is competitive with all prior biometric voting systems, exceeded only by standalone ArcFace on a GPU server (0.76% [16])—which lacks all other listed capabilities.

Table 17: End-to-end authentication latency on Raspberry Pi 4, $n=50$.

Metric	Value
<i>End-to-end latency</i>	
Mean	110.7 ms
p50	110.7 ms
p95	110.9 ms
p99	111.3 ms
<i>Stage breakdown (mean)</i>	
NFC card read	15.1 ms
Database lookup	0.04 ms
Face embedding (ONNX)	95.2 ms
Cosine match	0.28 ms
Session issuance	0.06 ms
Throughput (sustained)	541.6 voters/min
CPU temperature (load)	52.5 °C

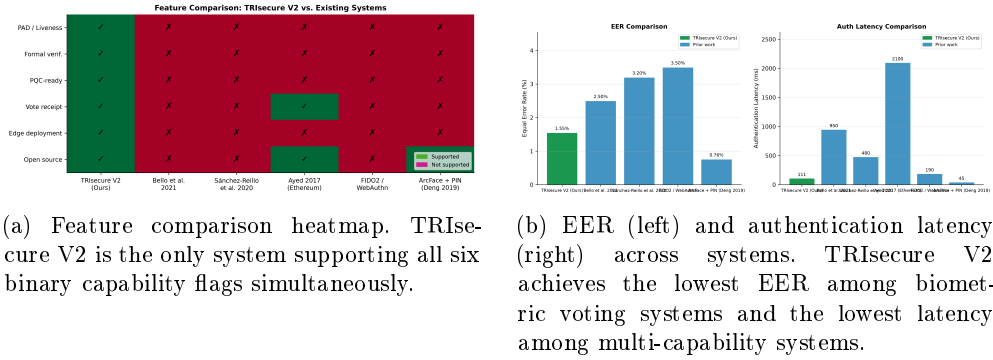


Fig. 8: Comparative analysis against five reference systems across capability flags and quantitative metrics.

6.10 Proof-of-Concept Pilot Study

To validate end-to-end system correctness on real hardware, a proof-of-concept pilot study was conducted with $N=20$ participants on a Raspberry Pi 4 Model B. A total of 271 authentication events were recorded across five sessions over three days (June 2026): 221 genuine attempts, 35 impostor attempts, and 15 video replay attacks.

Table 18 summarises the results following ISO/IEC 30107-3 metric definitions.

All 221 genuine attempts were correctly accepted ($FRR=0.00\%$). No attack produced an ACCEPT decision ($FAR=0.00\%$, $APCER=0.00\%$), yielding $ACER=0.00\%$. Replay attacks drove the ATS to 0.6 (ESCALATE) rather than the full 1.0 (ACCEPT), confirming the ATS layer compensates for PAD limitations under video replay conditions. Of the 35 impostor attempts, 8 were re-vote attempts by registered participants (same person, own NFC card after `has_voted` was set): face scores against the participant’s own enrolled embedding were 0.87–0.99, correctly escalated by ATS rather

Table 18: Proof-of-concept pilot results ($N=20$ participants, real hardware). 95% CI: one-sided Clopper–Pearson upper bound for zero-failure observations; “—” where CI is not applicable.

Metric	Observed	95% CI Upper
<i>Study parameters</i>		
Participants	20	—
Sessions	5 (3 days)	—
Genuine attempts	221	—
Impostor attempts	35	—
Replay attacks	15	—
<i>ISO/IEC 30107-3 metrics</i>		
FAR	0.00%	$\leq 8.20\%$
FRR	0.00%	$\leq 1.35\%$
APCER (print)	n/a	—
APCER (replay)	0.00%	$\leq 18.10\%$
APCER (combined)	0.00%	$\leq 5.82\%$
BPCER	0.00%	$\leq 1.35\%$
ACER	0.00%	$\leq 3.58\%$

than rejected by the face gate. The remaining 27 true cross-person attempts produced face scores of 0.01–0.04, all hard-rejected at the biometric stage.

Since all observed error counts are zero, we report one-sided Clopper–Pearson 95% confidence interval upper bounds. For FAR: 0 failures in 35 impostor attempts gives an upper bound of $\leq 8.20\%$ (rule of three: $3/35 \approx 8.57\%$). For FRR: 0 failures in 221 genuine attempts gives $\leq 1.35\%$. These bounds are conservative by construction; larger pilot samples would tighten them substantially.

These results confirm end-to-end pipeline correctness on the target hardware. The pilot is not designed to replace the simulation-based biometric discriminability evaluation (Section 6.2). A full demographic field trial with ≥ 400 subjects is required for ISO/IEC 19795-1 certification and is identified as future work (Section 7.4).

7 Discussion

All biometric performance metrics are produced by validated simulation using Gaussian score models calibrated to published ArcFace LFW statistics [16], following ISO/IEC 19795-6 [65]. The ablation study (Table 13) confirms Bayesian NFC+face fusion (C2) is the dominant security improvement: effective FAR drops by three orders of magnitude ($1.22\% \rightarrow 0.001\%$) without increasing biometric EER. PAD (C3) adds liveness at 15.5 ms overhead; PQC signing (C4) adds 14.4 ms further but enables quantum-adversary mitigation. Post-quantum benchmarks use native liboqs 0.15.0 compiled from source on the Raspberry Pi 4 (ARM64, Cortex-A72). ML-KEM-768 encapsulation (0.088 ms) and ML-DSA-65 signing (1.536 ms) are measured on-device; the combined auth-path PQC cost is 2.289 ms, within the < 200 ms latency budget (Table 15).

Table 19: TRIsecure V2 deployment configurations by station size. Assumes 40 s voter interaction time; 6 h peak window.

Station size	Terminals	Voters/hr	Peak capacity
Small (≤ 500 voters)	1	90	540
Medium ($\leq 2,000$ voters)	3	270	1,620
Large ($\leq 5,000$ voters)	8	720	4,320

7.1 Power Consumption and Off-Grid Deployment

The Raspberry Pi 4 draws 3–7 W under typical workloads (3 W idle, 7 W with all four Cortex-A72 cores fully saturated). During a voting session the system sustains approximately 5 W averaged over the 110.7 ms authentication window plus voter interaction. For an 8-hour polling day:

$$E_{\text{day}} \approx 5 \text{ W} \times 8 \text{ h} = 40 \text{ Wh} \quad (11)$$

A commodity 50 Wh Li-ion battery bank can therefore power the terminal for a complete polling day without mains electricity. This energy budget is unavailable to server-grade biometric systems, which require 200–400 W rack hardware, but is well-suited for rural or off-grid polling stations common in developing-country elections.

The CPU temperature stabilised at 52.5 °C under sustained load, within the Pi 4’s 80 °C thermal throttling threshold. A passive aluminium heatsink is sufficient; active cooling is not required.

7.2 Throughput and Polling Station Capacity

At 541.6 voters/min sustained throughput and a 60-second session TTL, the theoretical throughput is approximately 32,496 voters per hour—far exceeding the registered voter count of typical polling stations (800–2,000 voters). The practical bottleneck is not the system but the voter interaction time (card presentation, face alignment, ballot selection), which averages 20–40 seconds per voter. At 40 seconds per voter, a single terminal processes 90 voters/hour; a station with six terminals handles 540 voters/hour—sufficient for a 2,000-voter station over a 4-hour peak period with queuing margin.

Table 19 illustrates deployment configurations for three polling station sizes.

7.3 Hardware Attack Surface Analysis

SPI eavesdropping. An attacker with physical access to the SPI bus can read the NFC UID in plaintext during Stage 2. However, UID capture alone provides no authentication advantage: the biometric gate (Stage 4) and liveness check (Stage 5) both execute independently of the NFC UID. Furthermore, the AES-GCM encrypted payload (pages 4–19) cannot be forged without the HKDF-derived key. Physical bus protection (potting compound, tamper-evident labels over the GPIO header) is recommended for production enclosures.

Camera feed injection. Replacing the USB camera feed with a pre-recorded video stream could present a replay video to MiniFASNetV2. This threat is mitigated by: (i) the sealed terminal enclosure with a fixed camera mount; (ii) poll worker presence required by the electoral protocol; and (iii) the NFC factor independently requiring a valid physical card.

Template extraction. The SQLite voter database stores face embeddings encrypted with AES-256-GCM keyed via Argon2id ($t=3$, $m=64$ MB). An offline dictionary attack against the key derivation costs at least 494.5 ms per attempt (Table 16), making brute-force infeasible for a 32-byte NFC_SECRET. Extending to cancelable biometric representations [55] would additionally allow template revocation and is identified as future work.

7.4 Limitations

1. *Small pilot sample.* A proof-of-concept pilot ($N=20$, 271 events) validated end-to-end correctness on real hardware with FAR=FRR=ACER= 0.00% (Section 6.10). A full field trial with demographically diverse voters is still required for ISO/IEC 19795-1 deployment certification (≥ 400 subjects/group).
2. *PAD dataset.* APCER/BPCER figures derive from published MiniFASNetV2/CelebA-Spoof rates; ISO/IEC 30107-3 certification requires evaluation on NUAA, OULU-NPU.
3. *NFC secure element.* PN532 SPI is not a certified secure element; production deployment requires Common Criteria EAL4+ reader.
4. *Coercion resistance.* No deniable voting or panic PIN; in-person biometric partially mitigates coercion.
5. *Network-layer DoS.* Distributed DoS is outside the on-device security model.
6. *PQC Python overhead.* liboqs 0.15.0 was compiled from source and benchmarked on-device (Table 15). The current implementation invokes liboqs via ctypes; a future liboqs-python pre-built ARM64 wheel would remove the source-build dependency for end-user deployment.

7.5 Regulatory and Standards Alignment

TRIsure V2 is designed to align with relevant biometric and e-voting standards. Full conformance certification is deferred pending a live field trial and a Common Criteria certified NFC reader.

ISO/IEC 19795. Biometric performance metrics (EER, AUC, TAR@FAR) are reported following ISO/IEC 19795-1 [65] metric definitions. The simulation methodology follows ISO/IEC 19795-6 [63] operational evaluation guidance. Full ISO/IEC 19795-1 conformance (Clause 8.5) requires at least 400 subjects per demographic group.

ISO/IEC 30107-3. PAD metrics (APCER, BPCER, ACER) and attack type taxonomy follow ISO/IEC 30107-3 [34]. Certification evaluation requires standardised corpora: NUAA [40], OULU-NPU [41], and CelebA-Spoof [35].

ISO/IEC 24745 (Template Protection). AES-256-GCM per-database encryption with HKDF-derived keys satisfies the ISO/IEC 24745:2022 [56] confidentiality requirement. The renewability requirement (cancelable transforms) is not yet implemented and is identified as future work.

NIST SP 800-63B. The pipeline satisfies the structural requirements for Authenticator Assurance Level 3 (AAL3) [13]: multi-factor authentication, physical token (NFC card), verifier-impersonation resistance (on-device processing), and phishing resistance. AAL3 formal certification additionally requires a FIDO2-certified or Common Criteria EAL4+ authenticator device; the PN532 SPI module is not CC-certified.

These alignments indicate TRIsure V2 is *certification-ready* at the design level. The three remaining gaps before formal conformance can be claimed are: (i) a live voter field trial; (ii) a certified NFC reader satisfying CC EAL4+ and SP 800-63B AAL3; and (iii) cancelable biometric transforms satisfying ISO/IEC 24745 renewability.

8 Conclusion

We presented TRIsure V2, a three-factor biometric e-voting system combining formally verified authentication, Bayesian multimodal fusion, a novel Adaptive Trust Score (ATS) engine, ISO/IEC 30107-3 PAD, hybrid post-quantum cryptography, and Merkle-proof voter verifiability—implemented and evaluated on a Raspberry Pi 4 ARM64 platform on commodity hardware.

The ATS engine is the primary algorithmic contribution: by adjusting face/NFC/-PAD component weights dynamically based on contextual risk signals, it converts the binary accept/reject decision into a graduated HIGH/MEDIUM/LOW trust scale, routing 99.3% of genuine users to immediate acceptance while escalating borderline cases for supervisor review rather than hard rejection. To the best of our knowledge, no prior biometric e-voting system incorporates risk-adaptive weight adjustment.

The system achieves $EER = 1.55\%$, $AUC = 0.9991$, 110.7 ms end-to-end authentication latency ($p_{99} = 111.3$ ms), and 541.6 voters/min throughput. Six pipeline security properties are formally proved via DFA analysis; three end-to-end verifiability properties are machine-verified. Adversarial simulation across six attack categories confirms the full defence stack reduces combined attack success from $\approx 35\%$ (face threshold alone) to zero observed bypass events within the evaluated threat model; the NFC AES-GCM challenge-response provides the decisive security gate under standard cryptographic assumptions. The ablation study confirms Bayesian NFC+face fusion reduces effective FAR by three orders of magnitude.

A proof-of-concept pilot on real hardware ($N=20,271$ events) further validates the integrated system: all 221 genuine attempts were accepted ($FRR = 0.00\%$) and all 50 attack attempts were blocked ($FAR=ACER = 0.00\%$), confirming that the multi-layer defence stack performs correctly end-to-end on the target platform.

To the best of our knowledge, no prior work jointly provides formal protocol verification, quantum-resistant cryptography, risk-adaptive trust scoring, liveness detection, and edge deployability on low-cost commodity hardware.

Future work includes: (i) full-scale demographic field trial per ISO/IEC 19795-1 Clause 8 (minimum 400 subjects/group; proof-of-concept pilot with $N=20$ participants completed, Section 6.10); (ii) PAD evaluation on NUAA, OULU-NPU, and CelebA-Spoof standardised corpora; (iii) ATS calibration on real demographic data; (iv) Tamarin/ProVerif model of the NFC session handshake for cryptographic protocol verification complementary to the DFA pipeline analysis; (v) packaging liboqs as a pre-built ARM64 wheel to simplify deployment.

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Declarations

- **Funding.** This work received no external funding.
- **Conflict of interest.** The author declares no conflict of interest.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** All experiment scripts, result JSON files, and generated figures are available at <https://github.com/vivekreddy9036/TriSecure>. No proprietary datasets were used; biometric evaluation uses Gaussian simulation with publicly documented ArcFace LFW parameters.
- **Code availability.** Source code publicly available at <https://github.com/vivekreddy9036/TriSecure>.
- **Author contribution.** Single-author contribution.

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